Cloud Computing Mini Project Report

Repository: Harikrishan04/cloudComputing-assignment

Scope: Teacher-student parser pipeline, container, Kubernetes deployment, and CI/CD evidence

1. Executive Summary

A 50-record teacher subset was generated and cleaned successfully because the full teacher run exceeded the available time and memory budget. Two Pythia-70M student versions were trained, containerised, deployed, load tested, updated, and rolled back.

Component	Outcome
Teacher subset	50/50 records retained; 0 rejects; 0 duplicates
Base model	0% validity and field accuracy
Teacher validation	100% valid JSON; 81% intent; 39% full-frame
Student validation	0% valid JSON on 50-record subset
NACL simulation	6/6 packets passed
Kubernetes	2 ready parser replicas

2. Data and Teacher Labelling

The worker was hardened with bounded generation, JSON brace extraction, per-record progress, repository-relative prompts, and a 240-second timeout. Two worker shards were intended, but practical constraints led to one completed 50-record subset.

Metric	Value
Teacher records	50
Clean records	50
Rejected records	0
Duplicate records	0
Model context	1024 tokens
Generation limit	80 tokens
Teacher temperature/seed	0 / 42

3. Training and Evaluation

Both versions were trained with three epochs on CPU. Seeds were 42 and 43. Training losses improved materially but the small data subset did not produce reliable student JSON output.

Version	Seed	Loss start	Loss end
v1	42	1.9856	0.5867
v2	43	1.8685	0.5978

Model	Validity	Intent	Date	Time	Place	Full frame
Base	0%	0%	0%	0%	0%	0%
Student	0%	0%	0%	0%	0%	0%
Teacher	100%	81%	83%	83%	78%	39%

4. Container and Local Runtime

The parser image was built as parser-service:v1 and measured 1.69 GB. A one-CPU, one-GiB run passed 50 requests with no application failures.

Container measurement	Result
Image size	1685482224 bytes (1.57 GiB)
Requests	50
Failures	0
RPS	0.915
Duration	54.661 seconds
Peak memory	692.9 MiB

5. Kubernetes, Update, and Recovery

The final deployment has two ready replicas on the image-bearing k3d agent node, Never pull policy, an ingress route, readiness probes, resource limits, and an HPA of 1–2 replicas. SeaweedFS runs externally on the host; Kubernetes uses a ClusterIP service plus an Endpoints resource pointing to 172.18.0.1:8333.

Check	Result
Replica readiness	2/2
Rolling-update v2	Completed in 36 seconds
v2 header test	4/4 successful, all X-Model-Version=v2
Replica headers	Two pod names observed
Rollback	Restored v1 successfully
Pod recovery	Deleted pod recreated and returned Ready
Cold starts	16.425 s, 21.391 s, 19.023 s

6. Load and Network Validation

Two load runs were recorded. The header-aware 40-second test returned 15/15 successful requests at 0.39 RPS after rollout/recovery operations. The separate v2 run returned 4/4 successful requests and observed the v2 header on both replicas.

Test	Requests	Failures	RPS	Median / Avg ms
Earlier 30-second run	16	2 (502 recovery)	0.574	3600 / 3195
Final 40-second run	15	0	0.390	8200 / 9497
v2 10-second run	4	0	0.510	5800 / 3893
NACL packet checks	6	0		

The NACL sample passed all six packet decisions. The subnet design and packet table are recorded in network/subnets.md and network/nacl.json.

7. Cost, Limitations, and CI/CD

Cost estimate. Two replicas cost 2 × (1 × INR 3.50 + 1 × INR 0.45) = INR 7.90/hour. At the final measured 0.39 RPS, the estimated cost is INR 5.63 per 1000 requests. Container-run cost at 0.915 RPS is approximately INR 2.40 per 1000 requests.

Limitations. The teacher subset is not representative of the complete 3000-command pool. The student's 0% validity means production parsing is unreliable. Disk pressure required image storage cleanup and scheduling pinning. The separate v2 model is trained but also should not be deployed without quality-gate recovery.

CI/CD. The workflow contains lint, pytest, model evaluation, quality gate, artifact transfer, release download, and secret scanning. Final local validation passed: Ruff, four pytest tests, and NACL simulation. GitHub publication remains contingent on repository authentication.